MPG-Former: Drift-Aware Multi-scale Prototype-Graph Transformer for Cloud Workload Forecasting

(a) Input Preparation & History Representation

1 Raw Workload Window



$X \in \mathbb{R}^{B \times 72 \times C}$

48 history | 24 future

future target masked

2 Target RevIN

$target_history \in \mathbb{R}^{B \times 48}$



μ_w, σ_w → Normalize Target Only

3 History Representation Generation



$\tilde{X}^{enc}$ → Linear Projection → + PE → Gated Static Context

Static context (optional) → Gated Static Context

$H_s \in \mathbb{R}^{B \times 48 \times 128}$

(b) Multi-Evidence Extraction

4 Temporal Evidence Branch



H_s → Non-stationary Self-Attention → Moving-average Decomposition → Short-term Temporal Encoder Conv3 + Conv5 + BiGRU → c_{time} $B \times 128$

$A = \text{softmax}(\tau QK^T/\sqrt{d} + \delta)$



trend | seasonal

$encoded \in \mathbb{R}^{B \times 48 \times 128}$

5 Multi-scale Prototype Branch



p=6
p=12
p=24
p=48

Time tokens + FFT tokens → Scale Tokens → Prototype Matching Softmax → c_{proto} $B \times 128$

Prototype Memory M

$future_padding \in \mathbb{R}^{B \times 24}$

6 Dynamic Graph Branch



target-history similarity top-k



$workload_adj$





$context_adj$

GCN / Message Passing → c_{graph} $B \times 128$

contrastive alignment

(c) Drift-aware Fusion, Decoder & Outputs

7 Drift-aware Router

c_{time}
c_{graph}
c_{proto}

Drift features
z-score | |Δ|
std | entropy

Router MLP + Softmax

$c_{fused} = w_t c_t + w_g c_g + w_p c_p$

8 Decoder & Prediction Head

Last 12 Hist. + future_padding 24 → Enhanced Memory → Transformer Decoder

$memory = X_E + \text{Proj}(c_{fused})$

9 Outputs

Prediction Head →  → q10/q50/q90 interval

$\hat{y} \in \mathbb{R}^{B \times 24}$

router weights | drift score | prototype weights adaptive graph